



The Second Wave Should Go Narrow

The first generation of radioligand therapy proved the modality works. The next generation will decide whether it earns a permanent seat in oncology — and the way to earn it is to treat fewer patients, more deliberately, against a harder standard of proof.

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I want this modality to win. That's the lens everything here comes from. I spent a decade helping bring targeted radiation to market — the Axumin launch at Blue Earth, then Locametz and Pluvicto at Novartis — and I have real respect for what that first wave accomplished. Novartis took on the hardest job in the category: proving, in real patients, that an entirely new treatment paradigm could work at all. That's not a small thing, and the people who built that case earned the credit they've gotten for it. It's exactly because I want this modality to keep winning that I think the second wave should do some things differently than the first.

Radioligand therapy is roughly where immunotherapy stood around 2012. The foundational question has been answered. Targeted radiation works: it delivers dose to tumor, it produces meaningful responses, and it does so with a tolerability profile most oncologists can live with. That question is closed. The one that replaces it is harder and more consequential — not does it work, but do we know how to use it well enough to prove it deserves to move earlier in the treatment sequence.

The field's instinct, Novartis's first, has been to answer that question by expanding: more patients, earlier lines, broader labels — reaching as many people as the data could plausibly support. It's a good instinct, and an admirable one. Nobody sets out to treat fewer patients when treating more is on the table, and the ambition to reach the widest population possible comes from the right place. I think it's also, on the evidence so far, an instinct worth pressure-testing before the rest of the field simply repeats it — and the first wave already shows us why, at no fault to the people who built it in good faith.

What the first wave actually proved — and what it skipped

The headline achievement of the first generation is real and substantial, and it belongs mostly to Novartis. Pluvicto became the first PSMA-targeted radioligand therapy approved by the FDA, and demand has grown steadily — quarterly sales moved past half a billion dollars in 2025 on the strength of the pre-taxane expansion.¹ The modality is commercially validated because one company was willing to go first. Patients are being treated today who wouldn't otherwise have had this option, and the rest of the field owes that pioneering work a real debt.

How that expansion was won matters more than the milestone itself, for anyone building on top of it. The pre-taxane approval rested on PSMAfore, a trial that roughly tripled the eligible patient population² by moving Pluvicto ahead of chemotherapy. The primary endpoint — radiographic progression-free survival — came in strong. The commercial logic was clean: prove you can move earlier, and the addressable market multiplies.

The harder question is what that trial was built to answer. PSMAfore compared Pluvicto against a change in androgen-receptor pathway inhibitor — a second ARPI — in patients who had already progressed on

one. It delivered a progression-free survival win. It did not deliver an overall survival win, and the design allowed control-arm patients to cross over once they progressed, which muddled the survival read from the start.

The first wave optimized for the size of the eligible population. It did not optimize for the specificity of the patient or the strength of the proof. Those are different goals, and the gap between them is where the trouble lives.

That gap is the part that should shape everything the second wave does.

A different read from Europe

In March 2026, Novartis withdrew its application to bring Pluvicto into the earlier, pre-chemotherapy setting in the European Union.

The reasons are worth stating precisely, because they aren't the reasons people assume. The EMA's committee signaled it would not support the expansion. The comparator in PSMAfore — a second ARPI — was considered inadequate therapy for patients at that stage of disease, which made the progression-free survival advantage difficult to interpret. And there was no demonstrated overall survival benefit.³ Novartis was careful to note, accurately, that the withdrawal didn't touch the drug's existing approvals, its safety, or its efficacy — a responsible thing to clarify, and also not the question this piece is trying to answer.

Here's the detail that matters most. A US regulator accepted a progression-free survival win against a weak comparator and roughly tripled the label. A European regulator looked at the same data and reached a different conclusion: that the comparison wasn't fair, and that survival benefit hadn't been shown. Same evidence. Two verdicts. The modality came up short in Europe because the proof, as designed, wasn't built to survive a demanding reviewer — not because it treated too many patients, a genuinely useful thing to learn this early, while it's still cheap to fix.

The common framing — including one I used to reach for myself — is that the first wave went "too broad," and that breadth alone diluted the modality. That's close, but imprecise, and the imprecise version leads to the wrong lesson. The failure wasn't volume as such. A modality can treat a large population well. The real gap had two linked halves, and they fed each other.

The first half is about who was studied. By defining the trial population broadly — in effect, PSMA-positive and appropriate to delay chemotherapy — PSMAfore enrolled a clinically heterogeneous group: patients with very different tumor biology, different aggressiveness, different likelihoods of deep response, all averaged into a single result. Heterogeneous populations regress toward a muddier mean. When a trial doesn't specify who it's actually trying to treat, the responders and the non-responders blur together, and the magnitude of benefit reported ends up smaller than the magnitude the drug is capable of in the patients it truly helps. The trial didn't only rest on a soft comparator. It diluted its own effect size by not being more

specific about the population — an understandable instinct, in service of reaching more patients, that ended up working against the very case it was trying to make.

That dilution showed up constantly in the field during the launch, not just in the statistics. The product performed — nobody in an exam room was arguing otherwise. But the benefit, delivered into a heterogeneous population, wasn't compelling enough on its own to make a physician walk away from a chemotherapy sequence they already trusted and understood. A diluted effect size doesn't only complicate a regulator's read. It complicates the exact conversation a field team has to win at the bedside, one oncologist at a time.

The second half is about what it was measured against. A muted effect size needs a weak comparator to still look like a win — and a second ARPI in patients who had already failed one is a weak comparator. So the two halves compound: broad selection softens the result, the softened result needs a soft control to clear the bar, and the soft control is exactly what a serious regulator refuses. Breadth of ambition produced narrowness of proof, on both axes. That inversion is the real lesson, and a forgivable one to have missed the first time through — nobody had gone this way before.

The distinction that matters

Going broad at the bedside — treating a wide range of eligible patients — is a clinical and commercial choice that can be entirely defensible, and often the right one.

Going broad in your evidence — designing trials to clear the lowest bar that unlocks the largest label — is what erodes a modality's standing over time. The first wave conflated the two with the best of intentions. The second wave has a real opportunity to separate them.

Why the second wave is built to go narrow

What makes this moment interesting is that the second-generation agents are engineered around a different premise — whether or not their developers have framed it this way.

Consider Blue Earth Therapeutics' lead candidate, 177Lu-rhPSMA-10.1. The Phase 1 data that read out through 2025 didn't lead with population size. It led with dose ratios: a mean tumor-to-salivary-gland ratio of 73 and a tumor-to-kidney ratio of 32, delivering proportionately higher absorbed dose to tumor than to the critical healthy organs that limit how much radiation you can give.⁵ The design thesis is that a cleaner therapeutic ratio lets you push tumor dose without pushing toxicity in step.

That's a narrow-strategy instrument. A better therapeutic ratio is an argument for treating the right people more effectively, not simply treating more of them. The second wave's central advantage is precision, and precision and breadth-of-label pull in different directions. You can't lead with "we deliver dose more selectively" and then define success as "we qualified the widest possible population." The instrument and

the strategy have to agree — and agreeing means two things actually have to change, not just get better-argued.

The first is patient selection. The first-wave question was essentially binary: is the tumor PSMA-positive? That's a gate, not a prediction — and a gate lets in a heterogeneous crowd whose blended result understates what the drug does in its best responders. The next selection paradigm has to integrate PSMA PET with markers of discordant and aggressive biology, identifying up front the tumors where deep, durable response is most likely. Fewer patients enter. More of them benefit meaningfully. And the reported effect size climbs, because the results stop being averaged against patients who were never going to respond. This is the single highest-leverage change available to the second wave: kinder to the patient spared a therapy that won't help, and the thing that makes the eventual trial result strong enough to defend on its own terms.

The second is dosing. Much of the first-wave dosing paradigm — fixed activity, spaced cycles, a marrow-protection mindset — was carried over conservatively rather than derived from first principles, a reasonable choice when the modality was still new and unproven. A cleaner therapeutic ratio removes that excuse. It lets the second wave front-load dose where early tumor control matters most, adapt regimens to response instead of a fixed schedule, and individualize delivery the way the biology actually calls for — but only inside a population defined tightly enough that the biology is understood in the first place. Selection and dosing aren't two separate fixes. They're the same fix, applied at two different points in the same trial.

None of it holds without the discipline the first wave didn't yet have the hindsight to build: running trials against a comparator you'd be proud of, and being willing to be measured on overall survival rather than progression delayed alone. A narrower, cleaner win against a real standard is worth more than a broad win against a soft one.

A narrow, unambiguous proof of benefit is the most valuable asset the second wave can build. It's also the only one that travels across every regulator, every payer, and every skeptical tumor board.

Future shaping: getting ahead of the verdict

Most strategy in this field is reactive — read the trial result, then build the messaging around whatever you got. Call the alternative future shaping: deciding, years out, what you want to be true about the modality, then making the design choices today that let that future actually arrive. Not forecasting what happens. Building the conditions that decide it.

Applied here, that means asking a different question than the first wave asked — not how large a label can this data unlock, but what will a demanding reviewer, a cost-conscious payer, and a cautious oncologist need to believe three years from now, and what evidence, designed today, earns that belief.

The second wave has one advantage the first wave never had: it gets to watch the first wave's verdict come in before committing its own pivotal trials. That's a real gift, handed to it by the company willing to go first and take the earlier risks. Not using it — chasing the same broad labels against the same soft comparators out of habit — would be a genuine shame, and an avoidable one.



The company that shapes the next phase of radioligand therapy will be the one that proves, cleanly and against a fair standard, that targeted radiation works dramatically better when it's aimed with discipline — not necessarily the one with the widest label. That proof is narrow by construction, and it's also the only thing that turns a promising therapy into a permanent pillar of oncology. None of it would be possible without the first wave — without Novartis proving, against real skepticism, that this modality could work in the first place. The second wave's job is to build on what the first wave already won, not to outdo it.

NOTES & SOURCES

1. Novartis Q3 2025 financial reporting (Form 6-K): Pluvicto quarterly net sales of USD 564 million, +46%, driven by sustained US demand following the pre-taxane mCRPC approval and continued ex-US access expansion in the post-taxane setting; described as the only FDA- and PMDA-approved PSMA-targeted radioligand therapy for use after one ARPI and before chemotherapy.
2. FDA expanded-indication approval, 28 March 2025, based on PSMAfore (NCT04689828); the pre-taxane label was reported to approximately triple the number of patients eligible for Pluvicto. PSMAfore enrolled ~468 PSMA-positive mCRPC patients who had progressed on one ARPI and were considered appropriate to delay taxane chemotherapy.
3. European Medicines Agency, withdrawal of the type II variation to extend Pluvicto into the pre-chemotherapy setting; application withdrawn 23 March 2026. Per the CHMP's provisional assessment: the ARPI comparator was not considered adequate therapy for patients at that disease stage, complicating interpretation of the radiographic progression-free survival advantage, and no overall survival benefit was demonstrated versus the comparator arm.
4. Novartis statement, 24 April 2026: withdrawal followed CHMP feedback on the PSMAfore control arm and was not related to the quality, efficacy, or safety of Pluvicto, and did not affect ongoing trials, existing approvals, or pending submissions inside or outside the EU.
5. Blue Earth Therapeutics, Phase 1 dosimetry results for ¹⁷⁷Lu-rhPSMA-10.1 (NCT05413850), reported March 2025 and presented at SNMMI, June 2025: mean tumor-to-salivary-gland ratio of 73 and tumor-to-kidney ratio of 32 across 13 mCRPC patients; data described by the company as comparing favorably to published first-generation PSMA-targeted radioligand therapy dosimetry.

This perspective reflects the author's independent analysis of publicly available information and does not represent the position of any company referenced. It is commentary on the direction of the modality, not clinical or investment guidance.